

Requirements Report – *Voltcast*

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1. Project Overview

The Voltcast: Electricity Demand Forecast Engine is an end-to-end machine learning pipeline that predicts hourly electricity demand for the PJM Interconnection region. It uses historical EIA energy data and engineered time-series features.

The system integrates automated and secure data ingestion, XGBoost time series modeling, and deployment into a single, scalable pipeline. It will provide actionable forecasts via a FastAPI backend and an interactive Streamlit dashboard.

2. Objectives

- Automate ingestion and preprocessing of EIA energy demand data (PJM region) and engineered time-series features.
 - Train and evaluate baseline models and an XGBoost forecasting model.
 - Deploy a FastAPI backend exposing prediction, metrics, and health check endpoints.
 - Provide a Streamlit dashboard for visualization of historical demand, forecasts, and performance metrics.
 - Containerize and deploy the system on a cloud infrastructure (e.g., AWS or equivalent).
 - Deliver complete documentation (including architecture overview and API guide) and a demo video for stakeholders.
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3. Scope

In Scope

- **Data ingestion pipeline** (EIA PJM data only).
 - Baseline and **XGBoost** forecasting models.
 - **FastAPI** backend service with `/predict`, `/metrics`, and `/health` endpoints.
 - **Streamlit** dashboard for interactive visualization.
 - **Dockerized deployment** to AWS Free Tier (or equivalent).
 - Documentation and demo materials.
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4. Deliverables

- **Finalized Forecasting Model:** A fully trained and serialized **XGBoost** model object ready for production use.
 - **Production API Service:** The **FastAPI** backend, running in a container, which exposes the `/predict` and monitoring endpoints.
 - **Interactive Visualization Dashboard:** The **Streamlit dashboard** providing real-time visualization of historical data, current demand, and the 24-48 hour forecasts.
 - **Cloud Deployment Artifact:** The **Dockerized system** deployed and running on the free-tier cloud infrastructure.
 - **Project Documentation Package:** A complete set of documentation (including README, architecture overview, and API guide).
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5. Requirements

Functional

- System must fetch and process weather and energy demand data daily.
- Forecasts must cover the next 24–48 hours.
- API must provide endpoints for predictions, metrics, and health checks.
- Dashboard must display historical demand and forecasts clearly.